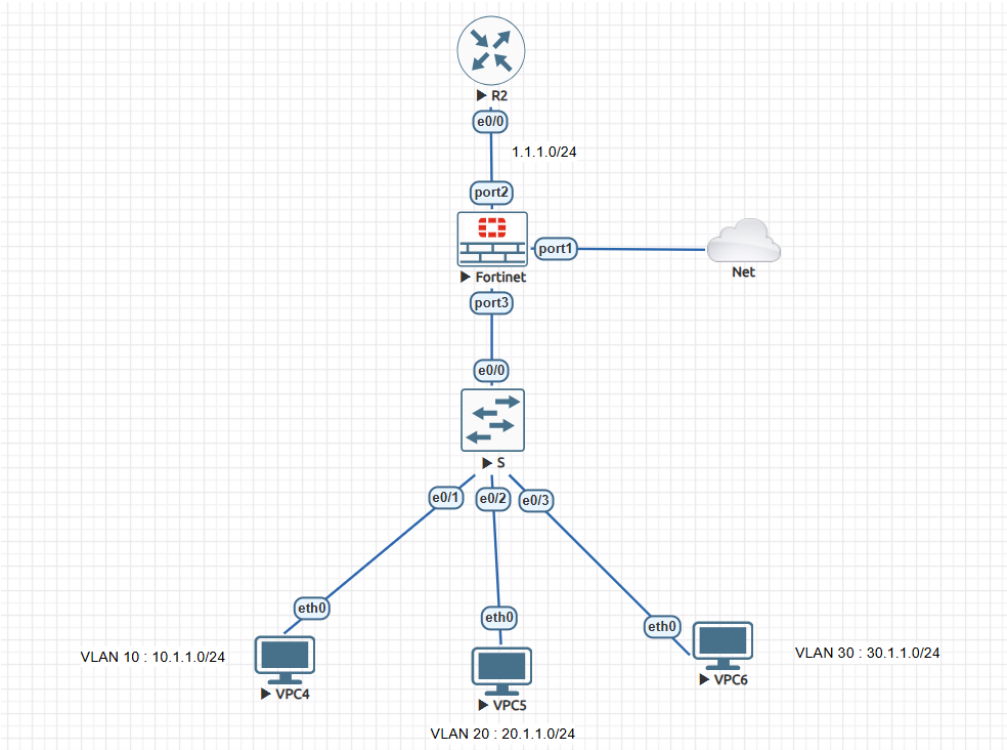
Objectifs

Mettre en avant le routage inter-VLAN (Inter-VLAN Routing) sur les Pare feux Fortigate. C'est même l'une de ses fonctions principales lorsqu'il est déployé en mode NAT (mode routeur). Dans une architecture d'entreprise, le FortiGate peut se positionner comme la passerelle par défaut de chaque VLAN, ce qui permet non seulement de router le trafic entre les différents segments réseau, mais aussi d'appliquer des politiques de sécurité et d'inspection à ces flux.

Topology



Router Configs

```
nternet(config)#do show ip int br
nterface                  IP-Address      OK? Method Status                Protocol
thernet0/0                1.1.1.1         YES manual up                    up
thernet0/1                unassigned      YES unset  administratively down down
thernet0/2                unassigned      YES unset  administratively down down
thernet0/3                unassigned      YES unset  administratively down down
oopback0                  8.8.8.8         YES manual up                    up
```

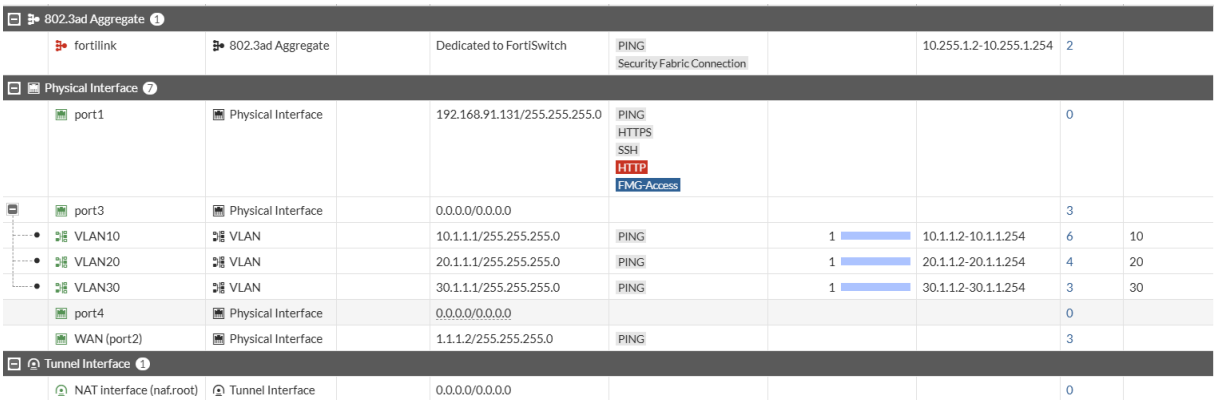
```
nternet(config)#do sh ip route
odes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
      a - application route
      + - replicated route, % - next hop override

ateway of last resort is not set

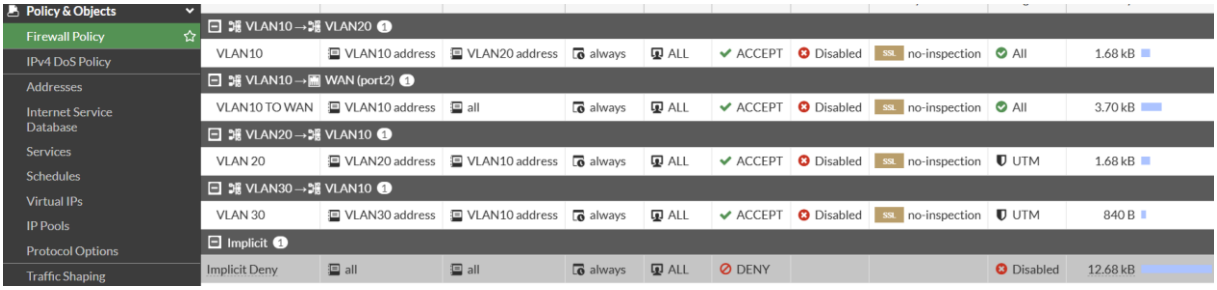
     1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
        1.1.1.0/24 is directly connected, Ethernet0/0
        1.1.1.1/32 is directly connected, Ethernet0/0
     8.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
        8.8.8.0/24 is directly connected, Loopback0
        8.8.8.8/32 is directly connected, Loopback0
     10.0.0.0/24 is subnetted, 1 subnets
        10.1.1.0 [110/11] via 1.1.1.2, 00:08:29, Ethernet0/0
     20.0.0.0/24 is subnetted, 1 subnets
        20.1.1.0 [110/11] via 1.1.1.2, 00:08:29, Ethernet0/0
     30.0.0.0/24 is subnetted, 1 subnets
        30.1.1.0 [110/11] via 1.1.1.2, 00:08:29, Ethernet0/0
```

```
router ospf 1
 router-id 1.1.1.1
 network 1.1.1.0 0.0.0.255 area 0.0.0.0
 network 8.8.8.8 0.0.0.3 area 0.0.0.0
```

Fortigate Configs



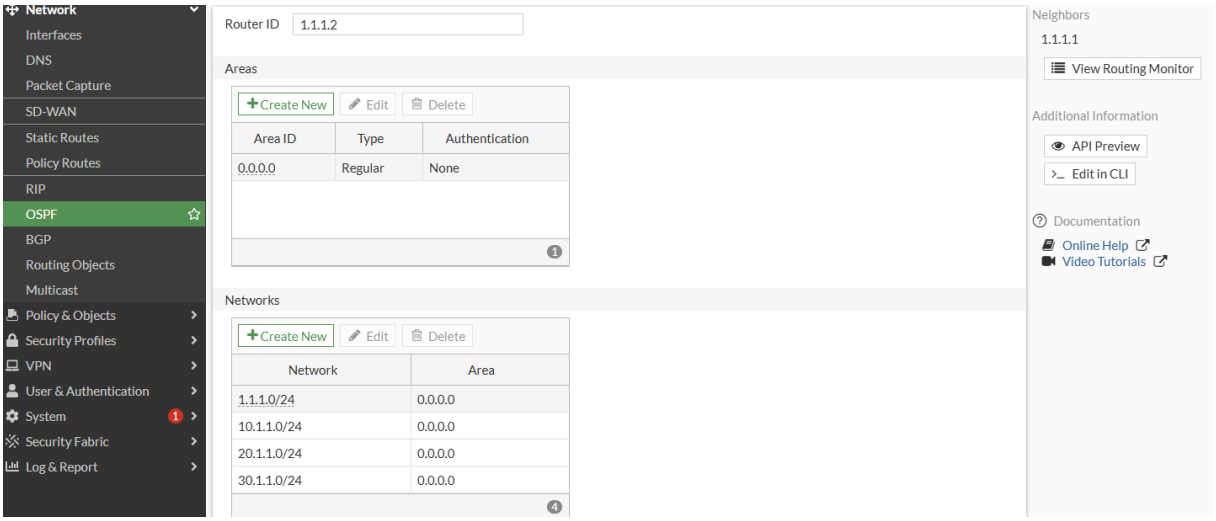
802.3ad Aggregate 1								
	fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2	
Physical Interface 7									
	port1	Physical Interface		192.168.91.131/255.255.255.0	PING HTTPS SSH HTTP FMG-Access			0	
	port3	Physical Interface		0.0.0.0/0.0.0.0				3	
	VLAN10	VLAN		10.1.1.1/255.255.255.0	PING	1	10.1.1.2-10.1.1.254	6	10
	VLAN20	VLAN		20.1.1.1/255.255.255.0	PING	1	20.1.1.2-20.1.1.254	4	20
	VLAN30	VLAN		30.1.1.1/255.255.255.0	PING	1	30.1.1.2-30.1.1.254	3	30
	port4	Physical Interface		0.0.0.0/0.0.0.0				0	
	WAN (port2)	Physical Interface		1.1.1.2/255.255.255.0	PING			3	
Tunnel Interface 1									
	NAT interface (naf.root)	Tunnel Interface		0.0.0.0/0.0.0.0				0	



Policy & Objects

- Firewall Policy
- IPv4 DoS Policy
- Addresses
- Internet Service Database
- Services
- Schedules
- Virtual IPs
- IP Pools
- Protocol Options
- Traffic Shaping

VLAN10 → VLAN20 1										
VLAN10	VLAN10 address	VLAN20 address	always	ALL	ACCEPT	Disabled	SSL no-inspection	All	1.68 kB	
VLAN10 → WAN (port2) 1										
VLAN10 TO WAN	VLAN10 address	all	always	ALL	ACCEPT	Disabled	SSL no-inspection	All	3.70 kB	
VLAN20 → VLAN10 1										
VLAN 20	VLAN20 address	VLAN10 address	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM	1.68 kB	
VLAN30 → VLAN10 1										
VLAN 30	VLAN30 address	VLAN10 address	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM	840 B	
Implicit 1										
Implicit Deny	all	all	always	ALL	DENY			Disabled	12.68 kB	



Ping Tests

```
VPCS> ping 20.1.1.2

84 bytes from 20.1.1.2 icmp_seq=1 ttl=63 time=2.346 ms
84 bytes from 20.1.1.2 icmp_seq=2 ttl=63 time=1.967 ms
84 bytes from 20.1.1.2 icmp_seq=3 ttl=63 time=2.133 ms
84 bytes from 20.1.1.2 icmp_seq=4 ttl=63 time=2.178 ms
84 bytes from 20.1.1.2 icmp_seq=5 ttl=63 time=0.928 ms

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=254 time=1.350 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=254 time=1.521 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=254 time=1.539 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=254 time=1.524 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=254 time=1.316 ms

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.1.1.2/24
GATEWAY     : 10.1.1.1
DNS         : 192.168.91.2
DHCP SERVER : 10.1.1.1
DHCP LEASE  : 601839, 604800/302400/529200
MAC         : 00:50:79:66:68:04
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

```
84 bytes from 10.1.1.1 icmp_seq=1 ttl=255 time=0.887 ms
84 bytes from 10.1.1.1 icmp_seq=2 ttl=255 time=1.298 ms
84 bytes from 10.1.1.1 icmp_seq=3 ttl=255 time=0.592 ms
84 bytes from 10.1.1.1 icmp_seq=4 ttl=255 time=1.138 ms
84 bytes from 10.1.1.1 icmp_seq=5 ttl=255 time=1.111 ms

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 20.1.1.2/24
GATEWAY     : 20.1.1.1
DNS         : 192.168.91.2
DHCP SERVER : 20.1.1.1
DHCP LEASE  : 604047, 604800/302400/529200
MAC         : 00:50:79:66:68:05
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

